Chapter 11. Fine-Tuning Representation Models for Classification

Supervised Classification

Fine-Tuning a Pretrained BERT Model

```
# Install specific versions of Hugging Face libraries
# to keep them compatible with each other.

!pip install \
transformers==4.46.3 \       # Main library for pretrained models like BERT
datasets==3.2.0 \            # Used to load and process datasets
huggingface_hub==0.26.2 \    # Used to download models and datasets from Hugging Face
tokenizers==0.20.3 \         # Used for fast text tokenization
accelerate==1.1.1            # Helps with efficient training on CPU/GPU
```

```
Requirement already satisfied: transformers==4.46.3 in /usr/local/lib/python3.12/dist-packages (4.46.3)
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  Using cached datasets-3.2.0-py3-none-any.whl.metadata (20 kB)
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Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch>=1
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Using cached datasets-3.2.0-py3-none-any.whl (480 kB)
Installing collected packages: datasets
  Attempting uninstall: datasets
    Found existing installation: datasets 2.18.0
    Uninstalling datasets-2.18.0:
      Successfully uninstalled datasets-2.18.0
Successfully installed datasets-3.2.0
```

```python
# Import the load_dataset function from the Hugging Face datasets library
from datasets import load_dataset

# Load the Rotten Tomatoes dataset
tomatoes = load_dataset("rotten_tomatoes")

# Separate the dataset into training and test sets
# Get the training part of the dataset
train_data = tomatoes["train"]

# Get the testing part of the dataset
test_data = tomatoes["test"]
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
```

```
# Import AutoTokenizer to convert text into tokens and token IDs
# Import AutoModelForSequenceClassification to load BERT
    # with a classification layer for predicting classes

from transformers import (
    AutoTokenizer,
    AutoModelForSequenceClassification
)

# Select the pretrained BERT model
model_id = "bert-base-cased"


# Load the pretrained BERT model
# AutoModelForSequenceClassification adds a classification head
    # on top of BERT so it can classify the input text
# num_labels=2 means we have two possible classes:
# 0 = Negative
# 1 = Positive
model = AutoModelForSequenceClassification.from_pretrained(
    model_id,
    num_labels=2
)

# Load the tokenizer suitable for this BERT model
tokenizer = AutoTokenizer.from_pretrained(model_id)
```

```
Some weights of BertForSequenceClassification were not initialized from the model checkpoint at bert-base-cased and are newl
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.
```

```
# DataCollatorWithPadding =  It automatically adds padding to the input sequences.
from transformers import DataCollatorWithPadding

# Create a data collator that dynamically adds padding
# to make sequences in each batch the same length
data_collator = DataCollatorWithPadding(

       # Use the same tokenizer that we loaded for BERT.
    # The tokenizer knows which padding token to use.
    tokenizer=tokenizer
)
```

```
def preprocess_function(examples):
    # Take the text from the dataset
    # Tokenize the text
    # Truncate sequences that are too long
    return tokenizer(
        examples["text"],
        truncation=True
    )
```

```
# Apply the preprocessing function to the training dataset
# batched=True means multiple examples are processed together
tokenized_train = train_data.map(
    preprocess_function,
    batched=True
)
```

Map: 100% 8530/8530 [00:01<00:00, 9249.88 examples/s]

```
# Apply the same preprocessing to the test dataset
tokenized_test = test_data.map(
    preprocess_function,
    batched=True
)
```

```
# upgrade the required datasets and scikit-learn versions
!pip install -U "datasets==2.18.0" "scikit-learn==1.3.2"
```

```
Collecting datasets==2.18.0
  Using cached datasets-2.18.0-py3-none-any.whl.metadata (20 kB)
Requirement already satisfied: scikit-learn==1.3.2 in /usr/local/lib/python3.12/dist-packages (1.3.2)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from datasets==2.18.0) (3.29.7)
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Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas->dat
Using cached datasets-2.18.0-py3-none-any.whl (510 kB)
Installing collected packages: datasets
  Attempting uninstall: datasets
    Found existing installation: datasets 3.2.0
    Uninstalling datasets-3.2.0:
      Successfully uninstalled datasets-3.2.0
Successfully installed datasets-2.18.0
```

```
!pip uninstall -y datasets
!pip install datasets==2.18.0
```

```
Found existing installation: datasets 2.18.0
Uninstalling datasets-2.18.0:
  Successfully uninstalled datasets-2.18.0
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Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets==2.18.0) (2025
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Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas->dat
Using cached datasets-2.18.0-py3-none-any.whl (510 kB)
Installing collected packages: datasets
Successfully installed datasets-2.18.0
```

```
# Import NumPy for numerical operations
import numpy as np

# Import the function used to load evaluation metrics
from datasets import load_metric

 # Function to calculate the F1 score
def compute_metrics(eval_pred):

    # Get the model's raw scores and the correct labels
    logits, labels = eval_pred

    # Select the class with the highest score
    predictions = np.argmax(logits, axis=-1)

    # Load the F1 metric
    load_f1 = load_metric("f1")

    # Compare predictions with the correct labels
    # predictions = what the model predicted
    # references = the correct answers from the dataset

    f1 = load_f1.compute(
        predictions=predictions,
        references=labels
    )["f1"]       # Get only the F1 score from the result

    # Return the F1 score
    return {"f1": f1}
```

```
from transformers import TrainingArguments

# Define how the model should be trained
training_args = TrainingArguments(
    "model",                         # Folder to save the model/checkpoints
    learning_rate=2e-5,              # Size of weight updates
    per_device_train_batch_size=16, # 16 examples per training batch
    per_device_eval_batch_size=16,  # 16 examples per evaluation batch
    num_train_epochs=1,              # Go through training data once
    weight_decay=0.01,               # Helps reduce overfitting
    save_strategy="epoch",           # Save after each epoch
    report_to="none"                 # Don't report to external services
)
```

```
# Import Trainer to handle the training and evaluation process
from transformers import Trainer

# Create the Trainer and give it everything needed for training
trainer = Trainer(

    # The BERT model that will be trained
    model=model,

     # Training settings such as learning rate, batch size, and epochs
    args=training_args,

    # Tokenized training data used to train the model
    train_dataset=tokenized_train,

    # Tokenized test data used to evaluate the model
    eval_dataset=tokenized_test,

     # BERT tokenizer used to convert text into tokens
    tokenizer=tokenizer,

     # Dynamically pads the sequences in each batch
    data_collator=data_collator,

     # Function used to calculate the F1 score during evaluation
    compute_metrics=compute_metrics,
)
```

```
/tmp/ipykernel_6765/980229885.py:4: FutureWarning: `tokenizer` is deprecated and will be removed in version 5.0.0 for `Trair
  trainer = Trainer(
```

```
# Start the actual training/fine-tuning process
trainer.train()
```

```
/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py:775: UserWarning: 'pin_memory' argument is set as tru
  super().__init__(loader)
```

[356/534 55:51 < 28:05, 0.11 it/s, Epoch 0.66/1]

Step	Training Loss

[534/534 1:25:50, Epoch 1/1]

Step	Training Loss
500	0.412400

```
TrainOutput(global_step=534, training_loss=0.4086745204996973, metrics={'train_runtime': 5165.7434,
'train_samples_per_second': 1.651, 'train_steps_per_second': 0.103, 'total_flos': 227605451772240.0, 'train_loss':
```

```
# Evaluate the trained model on the test dataset
trainer.evaluate()
```

```
/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py:775: UserWarning: 'pin_memory' argument is set as tru
  super().__init__(loader)
```

[67/67 03:07]

```
/tmp/ipykernel_42494/3220952111.py:12: FutureWarning: load_metric is deprecated and will be removed in the next major versic
  load_f1 = load_metric("f1")
/usr/local/lib/python3.12/dist-packages/datasets/load.py:756: FutureWarning: The repository for f1 contains custom code whic
You can avoid this message in future by passing the argument `trust_remote_code=True`.
Passing `trust_remote_code=True` will be mandatory to load this metric from the next major release of `datasets`.
  warnings.warn(
{'eval_loss': 0.3673253059387207,
 'eval_f1': 0.8460076045627377,
 'eval_runtime': 190.1179,
 'eval_samples_per_second': 5.607,
 'eval_steps_per_second': 0.352,
 'epoch': 1.0}
```

Freezing Layers

```
# Import the tokenizer and the model class for sequence classification
from transformers import (
    AutoTokenizer,
    AutoModelForSequenceClassification
)

# Select the pretrained BERT model
model_id = "bert-base-cased"

# Load BERT and add a classification head
# There are 2 classes: Positive and Negative
model = AutoModelForSequenceClassification.from_pretrained(
    model_id,
    num_labels=2
)

# Load the tokenizer suitable for this BERT model
tokenizer = AutoTokenizer.from_pretrained(model_id)
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
Some weights of BertForSequenceClassification were not initialized from the model checkpoint at bert-base-cased and are newl
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.
```

```
# Print the names of all layers and parameters in the BERT model
for name, param in model.named_parameters():
  print(name)
```

```
bert.encoder.layer.9.attention.output.LayerNorm.weight
bert.encoder.layer.9.attention.output.LayerNorm.bias
bert.encoder.layer.9.intermediate.dense.weight
bert.encoder.layer.9.intermediate.dense.bias
bert.encoder.layer.9.output.dense.weight
bert.encoder.layer.9.output.dense.bias
bert.encoder.layer.9.output.LayerNorm.weight
bert.encoder.layer.9.output.LayerNorm.bias
bert.encoder.layer.10.attention.self.query.weight
bert.encoder.layer.10.attention.self.query.bias
bert.encoder.layer.10.attention.self.key.weight
bert.encoder.layer.10.attention.self.key.bias
bert.encoder.layer.10.attention.self.value.weight
bert.encoder.layer.10.attention.self.value.bias
bert.encoder.layer.10.attention.output.dense.weight
bert.encoder.layer.10.attention.output.dense.bias
bert.encoder.layer.10.attention.output.LayerNorm.weight
bert.encoder.layer.10.attention.output.LayerNorm.bias
bert.encoder.layer.10.intermediate.dense.weight
bert.encoder.layer.10.intermediate.dense.bias
bert.encoder.layer.10.output.dense.weight
bert.encoder.layer.10.output.dense.bias
bert.encoder.layer.10.output.LayerNorm.weight
bert.encoder.layer.10.output.LayerNorm.bias
bert.encoder.layer.11.attention.self.query.weight
bert.encoder.layer.11.attention.self.query.bias
bert.encoder.layer.11.attention.self.key.weight
bert.encoder.layer.11.attention.self.key.bias
bert.encoder.layer.11.attention.self.value.weight
bert.encoder.layer.11.attention.self.value.bias
bert.encoder.layer.11.attention.output.dense.weight
bert.encoder.layer.11.attention.output.dense.bias
bert.encoder.layer.11.attention.output.LayerNorm.weight
bert.encoder.layer.11.attention.output.LayerNorm.bias
bert.encoder.layer.11.intermediate.dense.weight
bert.encoder.layer.11.intermediate.dense.bias
bert.encoder.layer.11.output.dense.weight
bert.encoder.layer.11.output.dense.bias
bert.encoder.layer.11.output.LayerNorm.weight
bert.encoder.layer.11.output.LayerNorm.bias
bert.pooler.dense.weight
bert.pooler.dense.bias
classifier.weight
classifier.bias
```

```
# Go through every parameter in the BERT model
for name, param in model.named_parameters():

# Check whether the parameter belongs to the classification head
 if name.startswith("classifier"):

  # Allow the classification head to be updated during training
  param.requires_grad = True

  # If the parameter is not part of the classification head
else:
   # Freeze that parameter so its values do not change during training
  param.requires_grad = False
```

```
from transformers import TrainingArguments, Trainer
# Trainer which executes the training process
trainer = Trainer(
model=model,
args=training_args,
train_dataset=tokenized_train,
eval_dataset=tokenized_test,
tokenizer=tokenizer,
data_collator=data_collator,
compute_metrics=compute_metrics,
)
trainer.train()
```

```
/tmp/ipykernel_6367/402954273.py:3: FutureWarning: `tokenizer` is deprecated and will be removed in version 5.0.0 for `Trair
  trainer = Trainer(
```

[534/534 01:58, Epoch 1/1]

Step	Training Loss
500	0.402500

```
TrainOutput(global_step=534, training_loss=0.39962195099962783, metrics={'train_runtime': 120.5573,
'train_samples_per_second': 70.755, 'train_steps_per_second': 4.429, 'total_flos': 227605451772240.0, 'train_loss':
```

```
trainer.evaluate()
```

[67/67 00:03]

```
/tmp/ipykernel_6367/3220952111.py:12: FutureWarning: load_metric is deprecated and will be removed in the next major versior
  load_f1 = load_metric("f1")
/usr/local/lib/python3.12/dist-packages/datasets/load.py:756: FutureWarning: The repository for f1 contains custom code whic
You can avoid this message in future by passing the argument `trust_remote_code=True`.
Passing `trust_remote_code=True` will be mandatory to load this metric from the next major release of `datasets`.
  warnings.warn(
```

Downloading builder script: 6.50k/? [00:00<00:00, 139kB/s]

```
{'eval_loss': 0.3667151629924774,
 'eval_f1': 0.8552754435107377,
 'eval_runtime': 4.8885,
 'eval_samples_per_second': 218.065,
 'eval_steps_per_second': 13.706,
 'epoch': 1.0}
```

```
Freezing first 10 encoder blocks of BERT model
```

```
# Import the tokenizer and the model class for sequence classification
from transformers import (
    AutoTokenizer,
    AutoModelForSequenceClassification
)

# Select the pretrained BERT model
model_id = "bert-base-cased"

# Load BERT and add a classification head
# There are 2 classes: Positive and Negative
model = AutoModelForSequenceClassification.from_pretrained(
    model_id,
    num_labels=2
)

# Load the tokenizer suitable for this BERT model
tokenizer = AutoTokenizer.from_pretrained(model_id)
```

```
Some weights of BertForSequenceClassification were not initialized from the model checkpoint at bert-base-cased and are newl
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.
```

```
# Go through all model parameters with their index and name
for index, (name, param) in enumerate(model.named_parameters()):

  # Freeze all parameters before index 165
  # This means the earlier BERT layers will not be updated during training

  if index < 165:

    # Stop these parameters from learning/updating
    param.requires_grad = False
```

```
# Trainer which executes the training process
trainer = Trainer(
model=model,
args=training_args,
train_dataset=tokenized_train,
eval_dataset=tokenized_test,
tokenizer=tokenizer,
data_collator=data_collator,
compute_metrics=compute_metrics,
)
```

```
/tmp/ipykernel_6367/2008171432.py:2: FutureWarning: `tokenizer` is deprecated and will be removed in version 5.0.0 for `Trai
  trainer = Trainer(
```

```
trainer.train()
```

[534/534 00:52, Epoch 1/1]

Step	Training Loss
500	0.476100

```
TrainOutput(global_step=534, training_loss=0.4725381408291363, metrics={'train_runtime': 52.7971,
'train_samples_per_second': 161.562, 'train_steps_per_second': 10.114, 'total_flos': 227605451772240.0, 'train_loss':
```

```
trainer.evaluate()
```

```
[67/67 00:03]
/usr/local/lib/python3.12/dist-packages/datasets/load.py:756: FutureWarning: The repository for f1 contains custom code whic
You can avoid this message in future by passing the argument `trust_remote_code=True`.
Passing `trust_remote_code=True` will be mandatory to load this metric from the next major release of `datasets`.
  warnings.warn(
{'eval_loss': 0.4100644588470459,
 'eval_f1': 0.8072519083969466,
 'eval_runtime': 4.8963,
 'eval_samples_per_second': 217.714,
 'eval_steps_per_second': 13.684,
 'epoch': 1.0}
```

Few-Shot Classification

SetFit: Efficient Fine-Tuning with Few Training Examples

```
 # Install the SetFit library to the latest compatible version
!pip install -U setfit
```

```
Requirement already satisfied: requests>=2.32.2 in /usr/local/lib/python3.12/dist-packages (from datasets>=2.15.0->setfit)
Requirement already satisfied: tqdm>=4.66.3 in /usr/local/lib/python3.12/dist-packages (from datasets>=2.15.0->setfit) (4.
Requirement already satisfied: xxhash in /usr/local/lib/python3.12/dist-packages (from datasets>=2.15.0->setfit) (3.8.1)
Requirement already satisfied: multiprocess<0.70.17 in /usr/local/lib/python3.12/dist-packages (from datasets>=2.15.0->set
Requirement already satisfied: fsspec<=2024.9.0,>=2023.1.0 in /usr/local/lib/python3.12/dist-packages (from fsspec[http]<=
Requirement already satisfied: aiohttp in /usr/local/lib/python3.12/dist-packages (from datasets>=2.15.0->setfit) (3.14.1)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from datasets>=2.15.0->setfit) (6.0
Requirement already satisfied: typing-extensions>=3.7.4.3 in /usr/local/lib/python3.12/dist-packages (from huggingface_hub
Requirement already satisfied: torch>=1.11.0 in /usr/local/lib/python3.12/dist-packages (from sentence-transformers>=3->se
Requirement already satisfied: scipy>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from sentence-transformers>=3->sen
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn->setfit) (1.5.3
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn->setfit)
Requirement already satisfied: accelerate>=0.20.3 in /usr/local/lib/python3.12/dist-packages (from sentence-transformers[t
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers>=4.41.0->se
Requirement already satisfied: tokenizers<0.21,>=0.20 in /usr/local/lib/python3.12/dist-packages (from transformers>=4.41.
Requirement already satisfied: safetensors>=0.4.1 in /usr/local/lib/python3.12/dist-packages (from transformers>=4.41.0->s
Requirement already satisfied: psutil in /usr/local/lib/python3.12/dist-packages (from accelerate>=0.20.3->sentence-transf
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp->datasets>
Requirement already satisfied: aiosignal>=1.4.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp->datasets>=2.15.0
Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp->datasets>=2.15.0->s
Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.12/dist-packages (from aiohttp->datasets>=2.15.
Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.12/dist-packages (from aiohttp->datasets>=2.1
Requirement already satisfied: propcache>=0.2.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp->datasets>=2.15.0
Requirement already satisfied: yarl<2.0,>=1.17.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp->datasets>=2.15.
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2-
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets>=2
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datas
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datas
Requirement already satisfied: setuptools<82 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-tran
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-tran
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-tr
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-transformer
Requirement already satisfied: cuda-toolkit==12.8.1 in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,c
Requirement already satisfied: cuda-bindings<13,>=12.9.4 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->s
Requirement already satisfied: nvidia-cudnn-cu12==9.19.0.56 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0
Requirement already satisfied: nvidia-cusparselt-cu12==0.7.1 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.
Requirement already satisfied: nvidia-nccl-cu12==2.28.9 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->se
Requirement already satisfied: nvidia-nvshmem-cu12==3.4.5 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->
Requirement already satisfied: triton==3.6.0 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-tran
Requirement already satisfied: nvidia-cublas-cu12==12.8.4.1.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolki
Requirement already satisfied: nvidia-cuda-runtime-cu12==12.8.90.* in /usr/local/lib/python3.12/dist-packages (from cuda-t
Requirement already satisfied: nvidia-cufft-cu12==11.3.3.83.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolki
Requirement already satisfied: nvidia-cufile-cu12==1.13.1.3.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolki
Requirement already satisfied: nvidia-cuda-cupti-cu12==12.8.90.* in /usr/local/lib/python3.12/dist-packages (from cuda-too
Requirement already satisfied: nvidia-curand-cu12==10.3.9.90.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolk
Requirement already satisfied: nvidia-cusolver-cu12==11.7.3.90.* in /usr/local/lib/python3.12/dist-packages (from cuda-too
Requirement already satisfied: nvidia-cusparse-cu12==12.5.8.93.* in /usr/local/lib/python3.12/dist-packages (from cuda-too
Requirement already satisfied: nvidia-nvjitlink-cu12==12.8.93.* in /usr/local/lib/python3.12/dist-packages (from cuda-tool
Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.8.93.* in /usr/local/lib/python3.12/dist-packages (from cuda-too
Requirement already satisfied: nvidia-nvtx-cu12==12.8.90.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[c
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets>=2
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets>=2.15.0->set
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets>=2.15.0->s
Requirement already satisfied: cuda-pathfinder~=1.1 in /usr/local/lib/python3.12/dist-packages (from cuda-bindings<13,>=12
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas->d
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch>=1
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->torch>=1.11.0->sen
```

Using the previously loaded dataset

```
# Import sample_dataset from SetFit
# It is used to select a small number of examples from each class
from setfit import sample_dataset
```

```
# We simulate a few-shot setting by sampling 16 examples per class
sampled_train_data = sample_dataset(tomatoes["train"],
num_samples=16)
```

```
/usr/local/lib/python3.12/dist-packages/setfit/data.py:154: DeprecationWarning: DataFrameGroupBy.apply operated on the grour
  df = df.apply(lambda x: x.sample(min(num_samples, len(x)), random_state=seed))
/usr/local/lib/python3.12/dist-packages/jupyter_client/session.py:203: DeprecationWarning: datetime.datetime.utcnow() is der
  return datetime.utcnow().replace(tzinfo=utc)
```

```
# SetFitModel is used to build and train SetFit models
from setfit import SetFitModel

# Load a pretrained SentenceTransformer model
# "all-mpnet-base-v2" converts sentences into meaningful numerical embeddings
# SetFit will use these embeddings to perform classification
model = SetFitModel.from_pretrained("sentence-transformers/all-mpnet-base-v2")
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/file_download.py:1132: FutureWarning: `resume_download` is deprecate
  warnings.warn(
model_head.pkl not found on HuggingFace Hub, initialising classification head with random weights. You should TRAIN this moc
```

```
# Import SetFit's TrainingArguments class
   # We rename it to SetFitTrainingArguments to make it clear that
   # these arguments are specifically for SetFit
from setfit import TrainingArguments as SetFitTrainingArguments

# SetFitTrainer is responsible for training and evaluating the SetFit model
from setfit import Trainer as SetFitTrainer

# Define training arguments
args = SetFitTrainingArguments(
num_epochs=3, # The number of epochs to use for contrastive learning
num_iterations=20  # The number of text pairs to generate
)

# Set eval_strategy using the value stored in evaluation_strategy
# This helps maintain compatibility between different SetFit versions
args.eval_strategy = args.evaluation_strategy
```

```
!pip install -q \
transformers==4.41.2 \
sentence-transformers==3.0.1 \
setfit==1.0.3 \
huggingface_hub==0.23.5 \
accelerate==0.31.0 \
peft==0.11.1
```

```
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 43.8/43.8 kB 2.5 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 9.1/9.1 MB 80.9 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 227.1/227.1 kB 25.0 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 75.9/75.9 kB 8.3 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 402.8/402.8 kB 37.0 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 309.4/309.4 kB 31.8 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 251.6/251.6 kB 26.1 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 3.6/3.6 MB 85.8 MB/s eta 0:00:00
ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is
diffusers 0.39.0 requires huggingface-hub<2.0,>=0.34.0, but you have huggingface-hub 0.23.5 which is incompatible.
gradio 6.20.0 requires huggingface-hub<2.0,>=1.2.0, but you have huggingface-hub 0.23.5 which is incompatible.
```

```
# Create the SetFit trainer
trainer = SetFitTrainer(

    # The pretrained SetFit mode
    model=model,

    # Training settings defined earlier
    args=args,

    # Small few-shot training dataset
train_dataset=sampled_train_data,

     # Test dataset used for evaluation
eval_dataset=test_data,

    # Use F1 score to measure performance

metric="f1"
)
```

```
# Training loop
trainer.train()
```

Map: 100% 32/32 [00:00<00:00, 519.41 examples/s]

```
***** Running training *****
  Num unique pairs = 1280
  Batch size = 16
  Num epochs = 3
  Total optimization steps = 240
/usr/local/lib/python3.12/dist-packages/notebook/notebookapp.py:191: SyntaxWarning: invalid escape sequence '\/'
  | |_| | '_ \/ _` / _` |  _/ -_)
/usr/local/lib/python3.12/dist-packages/jupyter_client/session.py:203: DeprecationWarning: datetime.datetime.utcnow() is dep
  return datetime.utcnow().replace(tzinfo=utc)
/usr/local/lib/python3.12/dist-packages/notebook/utils.py:280: DeprecationWarning: distutils Version classes are deprecated.
  return LooseVersion(v) >= LooseVersion(check)
wandb: (1) Create a W&B account
wandb: (2) Use an existing W&B account
wandb: (3) Don't visualize my results
wandb: Enter your choice: 3
wandb: You chose "Don't visualize my results"
wandb: Using W&B in offline mode.
wandb: W&B API key is configured. Use `wandb login --relogin` to force relogin
/usr/local/lib/python3.12/dist-packages/jupyter_client/session.py:203: DeprecationWarning: datetime.datetime.utcnow() is dep
  return datetime.utcnow().replace(tzinfo=utc)
/usr/local/lib/python3.12/dist-packages/wandb/analytics/sentry.py:269: DeprecationWarning: Read the `app_url` setting from t
  app_url = wandb.util.app_url(tags["base_url"])  # type: ignore[index]
/usr/local/lib/python3.12/dist-packages/jupyter_client/session.py:203: DeprecationWarning: datetime.datetime.utcnow() is dep
  return datetime.utcnow().replace(tzinfo=utc)
Tracking run with wandb version 0.28.0
W&B syncing is set to `offline` in this directory. Run `wandb online` or set WANDB_MODE=online to enable cloud syncing.
Run data is saved locally in /content/wandb/offline-run-20260804_175128-80m9tw85
View this run in the terminal with `wandb leet`
wandb: Detected [huggingface_hub.inference] in use.
/usr/local/lib/python3.12/dist-packages/google/protobuf/internal/well_known_types.py:178: DeprecationWarning: datetime.datet
  self.FromDatetime(datetime.datetime.utcnow())
wandb: Use W&B Weave for improved LLM call tracing. Install Weave with `pip install weave` then add `import weave` to the to
wandb: For more information, check out the docs at: https://weave-docs.wandb.ai
```

[240/240 01:15, Epoch 3/0]

Step	Training Loss

```
/usr/local/lib/python3.12/dist-packages/sklearn/linear_model/_logistic.py:451: DeprecationWarning: scipy.optimize: The `disp
  opt_res = optimize.minimize(
/usr/local/lib/python3.12/dist-packages/google/protobuf/internal/well_known_types.py:178: DeprecationWarning: datetime.datet
  self.FromDatetime(datetime.datetime.utcnow())
/usr/local/lib/python3.12/dist-packages/jupyter_client/session.py:203: DeprecationWarning: datetime.datetime.utcnow() is dep
  return datetime.utcnow().replace(tzinfo=utc)
```

```
# Evaluate the model on our test data
trainer.evaluate()
```

```
***** Running evaluation *****
/usr/local/lib/python3.12/dist-packages/jupyter_client/session.py:203: DeprecationWarning: datetime.datetime.utcnow() is dep
  return datetime.utcnow().replace(tzinfo=utc)
/usr/local/lib/python3.12/dist-packages/google/protobuf/internal/well_known_types.py:178: DeprecationWarning: datetime.datet
  self.FromDatetime(datetime.datetime.utcnow())
```

Downloading builder script: 6.79k/? [00:00<00:00, 238kB/s]

```
/usr/local/lib/python3.12/dist-packages/jupyter_client/session.py:203: DeprecationWarning: datetime.datetime.utcnow() is dep
  return datetime.utcnow().replace(tzinfo=utc)
/usr/local/lib/python3.12/dist-packages/google/protobuf/internal/well_known_types.py:178: DeprecationWarning: datetime.datet
  self.FromDatetime(datetime.datetime.utcnow())
{'f1': 0.8465804066543438}
```

Continued Pretraining with Masked Language Modeling

```
# Import the tokenizer and MLM model classes
from transformers import AutoTokenizer, AutoModelForMaskedLM

# MLM means the model learns to predict masked/missing words in a sentence
model = AutoModelForMaskedLM.from_pretrained("bert-base-cased")

# The tokenizer converts text into tokens that BERT can understand
tokenizer = AutoTokenizer.from_pretrained("bert-base-cased")
```

```
Some weights of the model checkpoint at bert-base-cased were not used when initializing BertForMaskedLM: ['bert.pooler.dense
- This IS expected if you are initializing BertForMaskedLM from the checkpoint of a model trained on another task or with ar
- This IS NOT expected if you are initializing BertForMaskedLM from the checkpoint of a model that you expect to be exactly
```

```
# Define a function to tokenize the input text
def preprocess_function(examples):
```

```
  # Take the text and convert it into tokens
    # truncation=True cuts the text if it is too long
  return tokenizer(examples["text"], truncation=True
)

# Apply the preprocessing function to the training dataset
tokenized_train = train_data.map(preprocess_function,

  # batched=True processes multiple examples at once
batched=True
)

# Remove the original sentiment labels
# MLM does not use the positive/negative labels
tokenized_train = tokenized_train.remove_columns("label")

tokenized_test = test_data.map(preprocess_function, batched=True
)

# Remove the original sentiment labels
# MLM does not use the positive/negative labels
tokenized_test = tokenized_test.remove_columns("label")
```

Map: 100% 1066/1066 [00:00<00:00, 9715.92 examples/s]

```
# Import the data collator used for Masked Language Modeling
from transformers import DataCollatorForLanguageModeling


# Create a data collator for masking tokens
data_collator = DataCollatorForLanguageModeling(

  # Use the BERT tokenizer to tokenize and mask the text
tokenizer=tokenizer,

   # Enable Masked Language Modeling
  # This tells the collator to randomly hide tokens
mlm=True,

   # Mask 15% of the tokens in each input
mlm_probability=0.15
)

# Training arguments for parameter tuning
training_args = TrainingArguments(
"model",
learning_rate=2e-5,
per_device_train_batch_size=16,
per_device_eval_batch_size=16,
num_train_epochs=10,
weight_decay=0.01,
save_strategy="epoch",
report_to="none"
)


# Create the Trainer to manage the MLM training process
trainer = Trainer(
model=model,
args=training_args,
train_dataset=tokenized_train,
eval_dataset=tokenized_test,
tokenizer=tokenizer,
data_collator=data_collator
)
```

```
# Save pre-trained tokenizer
tokenizer.save_pretrained("mlm")

# Train model
trainer.train()

# Save updated model
model.save_pretrained("mlm")
```

[5340/5340 24:17, Epoch 10/10]

Step	Training Loss
500	2.604200
1000	2.371300
1500	2.307000
2000	2.190800
2500	2.150200
3000	2.089500
3500	2.053400
4000	1.993100
4500	1.987000
5000	1.955500

```
# Import the pipeline function from Transformers
from transformers import pipeline

# Create a fill-mask pipeline
# This pipeline is used to predict a missing word represented by [MASK]
mask_filler = pipeline("fill-mask", model="bert-base-cased")

# Give BERT a sentence containing [MASK]
# BERT will predict possible words that can replace [MASK]
preds = mask_filler("What a horrible [MASK]!")

# Go through each prediction returned by the model
for pred in preds:

  # Print the complete sentence after replacing [MASK] with the predicted word
  print(f">>> {pred["sequence"]}")
```

```
Some weights of the model checkpoint at bert-base-cased were not used when initializing BertForMaskedLM: ['bert.pooler.dense
- This IS expected if you are initializing BertForMaskedLM from the checkpoint of a model trained on another task or with ar
- This IS NOT expected if you are initializing BertForMaskedLM from the checkpoint of a model that you expect to be exactly
>>> What a horrible idea!
>>> What a horrible dream!
>>> What a horrible thing!
>>> What a horrible day!
>>> What a horrible thought!
```

```
# Load the BERT model that we continued pretraining and saved in the "mlm" folder
# The "fill-mask" pipeline is used to predict a missing [MASK] token
mask_filler = pipeline("fill-mask", model="mlm")

# Give the model a sentence containing a [MASK] token
# The model predicts possible words that can replace [MASK]
preds = mask_filler("What a horrible [MASK]!")

# Go through each prediction returned by the model
for pred in preds:

   # Print the complete sentence after replacing [MASK] with the predicted word
  print(f">>> {pred["sequence"]}")
```

```
>>> What a horrible movie!
>>> What a horrible film!
>>> What a horrible mess!
>>> What a horrible comedy!
>>> What a horrible story!
```

```
# Import the model class used for text classification
from transformers import AutoModelForSequenceClassification

# Load the BERT model that we previously continued pretraining using MLM
# "mlm" is the folder containing our updated BERT model
# Add a classification head with 2 output classes: Positive and Negative

model = AutoModelForSequenceClassification.from_pretrained("mlm",
num_labels=2)

# Load the tokenizer that belongs to our continuously pretrained model
# We use the same tokenizer that was saved with the "mlm" model
tokenizer = AutoTokenizer.from_pretrained("mlm")
```

```
Some weights of BertForSequenceClassification were not initialized from the model checkpoint at mlm and are newly initialize
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.
```

Named-Entity Recognition

```
# Remove the currently installed versions of these Hugging Face libraries
!pip uninstall -y datasets huggingface_hub transformers

# Install specific versions to keep the libraries compatible with each other
!pip install datasets==3.6.0 transformers==4.46.3 huggingface_hub==0.26.2
```

```
Found existing installation: datasets 3.2.0
Uninstalling datasets-3.2.0:
  Successfully uninstalled datasets-3.2.0
Found existing installation: huggingface-hub 0.26.2
Uninstalling huggingface-hub-0.26.2:
  Successfully uninstalled huggingface-hub-0.26.2
Found existing installation: transformers 4.46.3
Uninstalling transformers-4.46.3:
  Successfully uninstalled transformers-4.46.3
Collecting datasets==3.6.0
  Downloading datasets-3.6.0-py3-none-any.whl.metadata (19 kB)
Collecting transformers==4.46.3
  Using cached transformers-4.46.3-py3-none-any.whl.metadata (44 kB)
Collecting huggingface_hub==0.26.2
  Using cached huggingface_hub-0.26.2-py3-none-any.whl.metadata (13 kB)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from datasets==3.6.0) (3.29.7)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from datasets==3.6.0) (1.26.4)
Requirement already satisfied: pyarrow>=15.0.0 in /usr/local/lib/python3.12/dist-packages (from datasets==3.6.0) (18.1.0)
Requirement already satisfied: dill<0.3.9,>=0.3.0 in /usr/local/lib/python3.12/dist-packages (from datasets==3.6.0) (0.3.8)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (from datasets==3.6.0) (2.2.2)
Requirement already satisfied: requests>=2.32.2 in /usr/local/lib/python3.12/dist-packages (from datasets==3.6.0) (2.32.4)
Requirement already satisfied: tqdm>=4.66.3 in /usr/local/lib/python3.12/dist-packages (from datasets==3.6.0) (4.67.3)
Requirement already satisfied: xxhash in /usr/local/lib/python3.12/dist-packages (from datasets==3.6.0) (3.8.1)
Requirement already satisfied: multiprocess<0.70.17 in /usr/local/lib/python3.12/dist-packages (from datasets==3.6.0) (0.70.
Requirement already satisfied: fsspec<=2025.3.0,>=2023.1.0 in /usr/local/lib/python3.12/dist-packages (from fsspec[http]<=20
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages (from datasets==3.6.0) (26.2)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from datasets==3.6.0) (6.0.3)
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers==4.46.3) (202
Requirement already satisfied: tokenizers<0.21,>=0.20 in /usr/local/lib/python3.12/dist-packages (from transformers==4.46.3)
Requirement already satisfied: safetensors>=0.4.1 in /usr/local/lib/python3.12/dist-packages (from transformers==4.46.3) (0.
Requirement already satisfied: typing-extensions>=3.7.4.3 in /usr/local/lib/python3.12/dist-packages (from huggingface_hub==
Requirement already satisfied: aiohttp!=4.0.0a0,!=4.0.0a1 in /usr/local/lib/python3.12/dist-packages (from fsspec[http]<=202
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->c
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets==3.6
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->dataset
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->dataset
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets==3.6
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets==3.6.0) (2025.
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets==3.6.0) (202
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4
Requirement already satisfied: aiosignal>=1.4.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1-
Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->fs
Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1
Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.6
Requirement already satisfied: propcache>=0.2.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1-
Requirement already satisfied: yarl<2.0,>=1.17.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas->dat
Downloading datasets-3.6.0-py3-none-any.whl (491 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 491.5/491.5 kB 15.4 MB/s eta 0:00:00
Using cached transformers-4.46.3-py3-none-any.whl (10.0 MB)
Using cached huggingface_hub-0.26.2-py3-none-any.whl (447 kB)
Installing collected packages: huggingface_hub, transformers, datasets
ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is
diffusers 0.39.0 requires huggingface-hub<2.0,>=0.34.0, but you have huggingface-hub 0.26.2 which is incompatible.
gradio 6.20.0 requires huggingface-hub<2.0,>=1.2.0, but you have huggingface-hub 0.26.2 which is incompatible.
Successfully installed datasets-3.6.0 huggingface_hub-0.26.2 transformers-4.46.3
```

```
# Load the CoNLL-2003 dataset, which is commonly used for Named-Entity Recognition (NER)
# trust_remote_code=True allows the dataset loading script to run
dataset = load_dataset("conll2003", trust_remote_code=True)
```

```
README.md:     12.3k/? [00:00<00:00, 794kB/s]
conll2003.py:     9.57k/? [00:00<00:00, 560kB/s]
Downloading data: 100%     983k/983k [00:00<00:00, 5.47MB/s]
Generating train split: 100%     14041/14041 [00:01<00:00, 9314.58 examples/s]
Generating validation split: 100%     3250/3250 [00:00<00:00, 9143.69 examples/s]
Generating test split: 100%     3453/3453 [00:00<00:00, 9480.61 examples/s]
```

```
# Select the 849th example from the training dataset
# Index 848 means we are selecting the example at position 848
example = dataset["train"][848]

example
```

```
{'id': '848',
 'tokens': ['Dean',
  'Palmer',
  'hit',
  'his',
  '30th',
  'homer',
  'for',
  'the',
  'Rangers',
  '.'],
 'pos_tags': [22, 22, 38, 29, 16, 21, 15, 12, 23, 7],
 'chunk_tags': [11, 12, 21, 11, 12, 12, 13, 11, 12, 0],
 'ner_tags': [1, 2, 0, 0, 0, 0, 0, 0, 3, 0]}
```

```
# Create a dictionary that maps each NER label to a unique number
label2id = {

    # O means the token is not a named entity
    "O": 0,

    # B-PER means the beginning of a person's name
    "B-PER": 1,

    # I-PER means inside/continuation of a person's name
    "I-PER": 2,

    # B-ORG means the beginning of an organization name
    "B-ORG": 3,

    # I-ORG means inside/continuation of an organization name
    "I-ORG": 4,

    # B-LOC means the beginning of a location name
    "B-LOC": 5,

    # I-LOC means inside/continuation of a location name
    "I-LOC": 6,

    # B-MISC means the beginning of a miscellaneous entity
    "B-MISC": 7,

    # I-MISC means inside/continuation of a miscellaneous entity
    "I-MISC": 8
}
```

```
# Create a reverse dictionary that maps each ID back to its NER label
id2label = {index: label
for label, index in label2id.items()}

# Display the original label-to-ID dictionary
label2id
```

```
{'O': 0,
 'B-PER': 1,
 'I-PER': 2,
 'B-ORG': 3,
 'I-ORG': 4,
 'B-LOC': 5,
 'I-LOC': 6,
 'B-MISC': 7,
 'I-MISC': 8}
```

```
# Import the model class used for token-level classification
from transformers import AutoModelForTokenClassification

# Load tokenizer
tokenizer = AutoTokenizer.from_pretrained("bert-base-cased")

# Load the pretrained BERT model with a token-classification head
model = AutoModelForTokenClassification.from_pretrained(

  # Use the pretrained BERT base cased model
"bert-base-cased",

   # Tell the model how many different NER labels it needs to predict
num_labels=len(id2label),

   # Tell the model which ID corresponds to which label
id2label=id2label,

  # Tell the model which label corresponds to which ID
label2id=label2id
)
```

```
Some weights of BertForTokenClassification were not initialized from the model checkpoint at bert-base-cased and are newly i
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.
```

```
# Define a function to tokenize the words and align their NER labels
def align_labels(examples):

    # Tokenize the words into BERT tokens/subwords
    # examples["tokens"] contains the words from the CoNLL-2003 dataset
    token_ids = tokenizer(
        examples["tokens"],
        truncation=True,
        is_split_into_words=True
    )

    # Get the original NER labels for each word
    # Example: [1, 2, 0, 0, 3]
    labels = examples["ner_tags"]

    # Create an empty list to store the new labels
    # These labels will match the BERT tokens
    updated_labels = []

    # Go through each sentence in the batch
    # index = sentence number
    # label = NER labels belonging to that sentence
    for index, label in enumerate(labels):

        # Get the original word number for every BERT token
        # This tells us which word each token came from
        word_ids = token_ids.word_ids(batch_index=index)

        # Keep track of the previous word number
        # We start with None because there is no previous word
        previous_word_idx = None

        # Create an empty list to store labels for this sentence
        label_ids = []

        # Go through every token in the sentence
        for word_idx in word_ids:

            # Check whether this token belongs to a new word
            if word_idx != previous_word_idx:

                # Remember the current word number
                previous_word_idx = word_idx

                # If word_idx is None, this is a special token such as [CLS] or [SEP]
                # Otherwise, get the original NER label of that word
                updated_label = -100 if word_idx is None else label[word_idx]

                # Add the label to the list
                label_ids.append(updated_label)

            # Handle special tokens such as [CLS] and [SEP]
            elif word_idx is None:

                # Use -100 so the model ignores these tokens during loss calculation
                label_ids.append(-100)
```

```
            # This token is another subtoken of the same word
            else:

                # Get the original NER label of the word
                updated_label = label[word_idx]

                # If the label is B-XXX, change it to I-XXX
                # This is needed because this token is inside the word,
                # not the beginning of the entity
                if updated_label % 2 == 1:
                    updated_label += 1

                # Add the updated label
                label_ids.append(updated_label)

        # Store the labels for this sentence
        updated_labels.append(label_ids)

    # Add the new token-level labels to the tokenized data
    token_ids["labels"] = updated_labels

    # Return the tokenized data with the aligned labels
    return token_ids


# Apply align_labels to the entire dataset
# batched=True means multiple sentences are processed together
tokenized = dataset.map(
    align_labels,
    batched=True
)
```

```
Map: 100%    14041/14041 [00:03<00:00, 5469.03 examples/s]
Map: 100%    3250/3250 [00:00<00:00, 7171.34 examples/s]
Map: 100%    3453/3453 [00:00<00:00, 7793.62 examples/s]
```

```
# Print the original NER labels from the dataset
print(f"Original: {example['ner_tags']}")

# Print the updated labels after tokenization and label alignment
print(f"Updated: {tokenized['train'][848]['labels']}")
```

```
Original: [1, 2, 0, 0, 0, 0, 0, 0, 3, 0]
Updated: [-100, 1, 2, 0, 0, 0, 0, 0, 0, 0, 3, 0, -100]
```

```
!pip install -U evaluate seqeval
```

```
Requirement already satisfied: evaluate in /usr/local/lib/python3.12/dist-packages (0.4.6)
Requirement already satisfied: seqeval in /usr/local/lib/python3.12/dist-packages (1.2.2)
Requirement already satisfied: datasets>=2.0.0 in /usr/local/lib/python3.12/dist-packages (from evaluate) (3.6.0)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from evaluate) (1.26.4)
Requirement already satisfied: dill in /usr/local/lib/python3.12/dist-packages (from evaluate) (0.3.8)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (from evaluate) (2.2.2)
Requirement already satisfied: requests>=2.19.0 in /usr/local/lib/python3.12/dist-packages (from evaluate) (2.32.4)
Requirement already satisfied: tqdm>=4.62.1 in /usr/local/lib/python3.12/dist-packages (from evaluate) (4.67.3)
Requirement already satisfied: xxhash in /usr/local/lib/python3.12/dist-packages (from evaluate) (3.8.1)
Requirement already satisfied: multiprocess in /usr/local/lib/python3.12/dist-packages (from evaluate) (0.70.16)
Requirement already satisfied: fsspec>=2021.05.0 in /usr/local/lib/python3.12/dist-packages (from fsspec[http]>=2021.05.0->e
Requirement already satisfied: huggingface-hub>=0.7.0 in /usr/local/lib/python3.12/dist-packages (from evaluate) (0.26.2)
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages (from evaluate) (26.2)
Requirement already satisfied: scikit-learn>=0.21.3 in /usr/local/lib/python3.12/dist-packages (from seqeval) (1.3.2)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from datasets>=2.0.0->evaluate) (3.29.7)
Requirement already satisfied: pyarrow>=15.0.0 in /usr/local/lib/python3.12/dist-packages (from datasets>=2.0.0->evaluate) (
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from datasets>=2.0.0->evaluate) (6.0.
Requirement already satisfied: aiohttp!=4.0.0a0,!=4.0.0a1 in /usr/local/lib/python3.12/dist-packages (from fsspec[http]>=202
Requirement already satisfied: typing-extensions>=3.7.4.3 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.19.0->e
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests>=2.19.0->evaluate) (3.
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests>=2.19.0->evaluat
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests>=2.19.0->evaluat
Requirement already satisfied: scipy>=1.5.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn>=0.21.3->seqeval)
Requirement already satisfied: joblib>=1.1.1 in /usr/local/lib/python3.12/dist-packages (from scikit-learn>=0.21.3->seqeval)
Requirement already satisfied: threadpoolctl>=2.0.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn>=0.21.3->s
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas->evaluate) (2.
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas->evaluate) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas->evaluate) (2026.3)
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4
Requirement already satisfied: aiosignal>=1.4.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1-
Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->fs
Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1
Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0
```

```
Requirement already satisfied: propcache>=0.2.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1-
Requirement already satisfied: yarl<2.0,>=1.17.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas->eva
```

```
import evaluate

# Load the evaluation library for Named Entity Recognition
seqeval = evaluate.load("seqeval")


def compute_metrics(eval_pred):

    # Separate model predictions and true labels
    logits, labels = eval_pred

    # Convert logits into predicted label IDs
    predictions = np.argmax(logits, axis=2)

    # Lists to store predicted and actual labels
    true_predictions = []
    true_labels = []

    # Loop through each sentence
    for prediction, label in zip(predictions, labels):

        # Loop through each token in the sentence
        for token_prediction, token_label in zip(prediction, label):

            # Ignore special tokens ([CLS], [SEP], padding)
            if token_label != -100:

                # Convert predicted ID to label name
                true_predictions.append(
                    [id2label[token_prediction]]
                )

                # Convert actual ID to label name
                true_labels.append(
                    [id2label[token_label]]
                )

    # Calculate evaluation metrics
    results = seqeval.compute(
        predictions=true_predictions,
        references=true_labels
    )

    # Return F1 score
    return {
        "f1": results["overall_f1"]
    }
```

Fine-Tuning for Named-Entity Recognition

```
from transformers import DataCollatorForTokenClassification

# Create a data collator for NER
# It dynamically pads the input tokens and their corresponding labels
data_collator = DataCollatorForTokenClassification(tokenizer=tokenizer)
```

```
# Training arguments for parameter tuning
training_args = TrainingArguments(
"model",
learning_rate=2e-5,
per_device_train_batch_size=16,
per_device_eval_batch_size=16,
num_train_epochs=1,
weight_decay=0.01,
save_strategy="epoch",
report_to="none"
)
```

```
# Initialize Trainer
trainer = Trainer(
model=model,
args=training_args,
train_dataset=tokenized["train"],
```